



YordamchiAI

Rasmiy Texnik Hujjatlar To'plami

Hujjat: MASTER PLATFORM AUDIT + ARCHITECTURE + ROADMAP

Versiya: v1.0.0 | **Sana:** 2026-06-25

Muallif: CTO Office — Arxitektura va Mahsulot Bo'limi

Holat: Konfidensial — Faqat Texnik Jamoa

Til: O'zbek (Latin) + Inglizcha texnik atamalar

AI-POWERED LEARNING PLATFORM

MUNDARIJA

1.	Executive Summary	3
2.	Texnologiya Stack Auditi	4
3.	Papka Tuzilmasi Auditi	5
4.	Sahifalar Auditi	6
5.	Komponentlar Auditi	8
6.	Database Auditi	10
7.	Autentifikatsiya Auditi	13
8.	Foydalanuvchi Profil Auditi	14
9.	AI Tizim Auditi	15
10.	Xavfsizlik Auditi	17
11.	Performance Auditi	18
12.	UI/UX Auditi	19
13.	Texnik Qarz	20
14.	Kelajak Arxitekturasi	22
15.	Rivojlanish Roadmap	24
16.	Xavf Tahlili	26
17.	TOP 100 Yaxshilanishlar	27
18.	Yakuniy Baho	30

1. EXECUTIVE SUMMARY

Platforma: YordamchiAI — AI-powered ta'lim platformasi (LMS)

Stack: React 19 + TypeScript 6 + Supabase + Google Gemini 2.5 Flash

Bosqich: MVP (Minimum Viable Product)

Umumiy baho: 5.3 / 10

YordamchiAI hozirda faol rivojlanish bosqichida bo'lgan AI-powered ta'lim platformasi. Platforma React 19 + Supabase stack'i asosida qurilgan bo'lib, asosiy AI engine sifatida Google Gemini 2.5 Flash ishlatiladi.

Kuchli Tomonlar

- **Supabase** — ajoyib Full-Stack BaaS tanlovi (Auth + DB + Storage + Edge Functions bitta joyda)
- **Row Level Security (RLS)** — barcha jadvallarda mavjud, yaxshi xavfsizlik asosi
- **AI Provider Abstraction** — model almashtirish uchun interface mavjud
- **AsomiddinAvatar Design System** — premium dizayn, production sifatida
- **TypeScript 6** — kuchli typing butun platformada
- **i18n** — uz/ru/en til qo'llab-quvvatlashi mavjud

Kritik Muammolar

● ZUDLIK BILAN HAL QILINISHI KERAK:

- Hech qanday test yo'q (Playwright o'rnatilgan, ammo ishlatilmaydi)
- Error boundary yo'q — bitta xato butun ilovani to'xtatadi
- Streaming haqiqiy emas — bu simulyatsiya
- Rate limiting yo'q — AI Edge Function himoyasiz
- Role selection xavfsizlik zaifligi (foydalanuvchi o'zi admin bo'la oladi)
- Code splitting yo'q — 1.37MB bitta bundle
- Accessibility deyarli nol (2/10)
- ProfileContext qurilgan, lekin hech qayerda ishlatilmaydi

6/10

Arxitektura

7/10

Database

7/10

UI Sifati

5/10

UX Sifati

4/10

Performance

5/10

Skalabillik

5/10

Maintainability

6/10

Xavfsizlik

7/10

AI Tayyorligi

2. TEXNOLOGIYA STACK AUDITI

Frontend Stack

Texnologiya	Versiya	Holat	Baho	Izoh
React	19.2.7	Ready	9/10	Eng so'nggi versiya, concurrent features
TypeScript	~6.0.2	Ready	8/10	Kuchli typing butun platformada
Vite	8.0.12	Ready	9/10	Juda tez build, HMR
Tailwind CSS	4.3.1	Improve	7/10	v4 yangi — plugin ekotizimi cheklangan
React Router	7.18.0	Ready	8/10	Yaxshi routing
Lucide React	1.20.0	Ready	8/10	Ko'p ikonkalar
jsPDF	4.2.1	Improve	5/10	Cheklangan, DB integration yo'q
Playwright	1.61.0	Missing	1/10	O'rnatilgan, LEKIN 0 ta test yozilmagan
Zustand / Redux	—	Missing	—	State management — kerak bo'ladi
TanStack Query	—	Missing	—	Server state caching — kerak bo'ladi

Backend Stack (Supabase)

Texnologiya	Holat	Baho	Izoh
PostgreSQL (Supabase)	Ready	9/10	Ishonchli RDBMS
Supabase Auth	Ready	8/10	JWT, session management

Supabase Storage	Improve	6/10	avatars bucket YO'Q — kritik
Edge Functions (Deno)	Ready	8/10	2 ta function: ai-chat, admin-users
Row Level Security	Ready	8/10	Barcha jadvallarda mavjud
Supabase Realtime	Missing	—	Haqiqiy streaming uchun kerak
pgvector	Missing	—	AI memory uchun kerak

AI Stack

Texnologiya	Holat	Baho	Izoh
Google Gemini 2.5 Flash	Ready	9/10	Eng yangi model, tez
Edge Function (ai-chat)	Improve	7/10	Rate limiting yo'q
True Streaming	Missing	—	Hozir setTimeout simulyatsiya
AI Memory / Embeddings	Missing	—	pgvector kerak
Context Management	Missing	—	Sliding window kerak
Token Tracking	Missing	—	Xarajat hisobi yo'q

3. MAVJUD XUSUSIYATLAR AUDITI

Autentifikatsiya Moduli

Xusuriyat	Holat
Email + Parol Login	To'liq
Ro'yxatdan o'tish	To'liq
Parol tiklash	Qisman
Remember Me	To'liq
Role-based routing	To'liq
Email tekshiruvi	Yo'q
OAuth (Google/GitHub)	Yo'q
2FA / MFA	Yo'q
CAPTCHA	Yo'q

AI Chat Moduli

Xusuriyat	Holat
Chat conversation	To'liq
Message persistence (Supabase)	To'liq
Gemini 2.5 Flash integration	To'liq
Student context injection	To'liq
Haqiqiy streaming	Yo'q
Simulyatsiya streaming	To'liq

Thinking indicator	To'liq
Markdown rendering	To'liq
Code blocks	To'liq
Fayl yuklash	Yo'q
Voice input	Yo'q

Talaba Paneli

Xusuriyat	Tayyor	Holat
Dashboard	45%	Qisman
Darslar ro'yxati	55%	Qisman
Davomat ko'rish	50%	Qisman
Testlar	55%	Qisman
Yutuqlar	60%	Qisman
AI Assistant	80%	Yaxshi
Profil sahifasi	35%	Kritik
Uy vazifasi	0%	Yo'q
Leaderboard	0%	Yo'q

O'qituvchi Paneli

Xusuriyat	Tayyor	Holat
Kurslar boshqaruvi	40%	Qisman
Guruh boshqaruvi	40%	Qisman
Davomat belgilash	45%	Qisman

Test yaratish	45%	 Qisman
Talaba progressi	0%	 Yo'q
Uy vazifasi berish	0%	 Yo'q
E'lonlar	0%	 Yo'q

4. DATABASE AUDITI

Mavjud Jadvallar (12 ta)

Jadval	Maqsad	RLS	Holat
profiles	Foydalanuvchi identifikatsiyasi	✓	● Yetishmaydi
subjects	Fanlar / kurslar	✓	✓ Yaxshi
groups	Sinf guruhlari	✓	● Soft delete yo'q
student_groups	Talaba-guruh M:N	✓	✓ Yaxshi
teacher_subjects	O'qituvchi-fan M:N	✓	● Kam ishlatiladi
lessons	Dars kontenti	✓	● Order index yo'q
lesson_attachments	Dars fayllari	✓	✓ Yaxshi
attendance	Davomat yozuvlari	✓	✓ Yaxshi
tests	Test ta'riflari (JSONB)	✓	● JSONB tekshiruv yo'q
test_results	Test natijalari	✓	● JSONB tekshiruv yo'q
achievement_definitions	Yutuq shablonlari	✓	● Qisman
user_achievements	Qo'lga kiritilgan yutuqlar	✓	● Qisman
user_score_snapshots	Ball tarixi	✓	● Qisman
settings	Global sozlamalar	✓	● Ishlatilmaydi
ai_conversations	AI chat sessiyalari	✓	✓ Yaxshi

Munosabatlar Diagrammasi

```
profiles (id)
|
|—[1:N]— student_groups (student_id → groups.id)
|         └─ groups (id)
|             └─ subjects (subject_id)
|                 └─ profiles (teacher_id) [O'qituvchi]
|
|—[M:N]— teacher_subjects (teacher_id → subjects.id)
|
|—[1:N]— attendance (student_id)
|       └─ groups (group_id)
|           └─ profiles (teacher_id)
|
|—[1:N]— test_results (student_id)
|       └─ tests (test_id)
|           └─ groups (group_id, nullable)
|               └─ subjects (subject_id, nullable)
|
|—[1:N]— user_achievements (profile_id)
|       └─ achievement_definitions (achievement_id)
|
|—[1:N]— user_score_snapshots (user_id)
|
|—[1:N]— ai_conversations (student_id)
|       └─[1:N]— ai_messages (conversation_id)

[GLOBAL] settings (key-value)

lessons (group_id → groups.id)
└─[1:N]— lesson_attachments (lesson_id)
```

Yetishmayotgan Kritik Jadvallar

Zudlik bilan kerak bo'ladiganlar:

- `homework` — Uy vazifalari
- `homework_submissions` — Uy vazifasi topshiriqlari
- `certificates` — Sertifikatlar (hozir faqat client-side PDF)
- `notifications` — Xabarnomalar

- `announcements` — E'lonlar
- `audit_logs` — Kim nima qilganini tracking

Kelajakda kerak bo'ladiganlar:

- `payments` , `subscriptions` — To'lov tizimi
- `comments` — Darslar uchun izohlar
- `leaderboard_snapshots` — Reyting tarixi
- `marketplace_items` — Marketplace
- `profiles.updated_at` — Hozir bu ustun YO'Q (kritik)

Storage Buckets

Bucket	Holat	Foydalanish
lesson-attachments	✓ Mavjud	Dars fayllari
avatars	✗ YO'Q — KRITIK	Avatar yuklash butunlay ishlamaydi

5. AI TIZIM AUDITI

Arxitektura

```
AIAssistantPage.tsx
├── EdgeFunctionProvider.complete()
│   └── supabase.functions.invoke('ai-chat')
│       ├── buildSystemPrompt(studentContext)
│       ├── Gemini 2.5 Flash API call
│       └── { response: text }
├── aiChatService.addMessage() → ai_messages (DB)
└── StreamingMessage (SIMULYATSIYA — haqiqiy emas)
```

Kuchli Tomonlar

- **AIProvider interface** — model almashtirish uchun abstraction mavjud (arxitektura jihatidan to'g'ri)
- **Student context injection** — guruhlar, darslar, testlar, davomat AI'ga yuboriladi
- **Thinking parts filtering** — Gemini 2.5 Flash "thinking" partlari filtrlanadi (kritik bug tuzatildi)
- **thinkingBudget: 0** — keraksiz token sarfi oldini oladi
- **Message persistence** — Supabase DB'da saqlanadi
- **Brand identity** — AsomiddinAvatar premium dizayn

Kritik Zaif Tomonlar

Muammo	Jiddiylilik	Ta'sir
Haqiqiy streaming yo'q (setTimeout simulyatsiya)	Yuqori	UX va texnik to'g'rilik
Rate limiting yo'q	Kritik	API kvota tez tugaydi
Token tracking yo'q	Kritik	Xarajatni nazorat qilib bo'lmaydi
Context window boshqarilmaydi	Yuqori	Token limit, yuqori xarajat

MockAIProvider bundleda	O'rta	Dead code, chalkashlik
AIAssistantPage.tsx 700+ qator	Yuqori	Maintainability
AI Prompt Injection himoyasi yo'q	O'rta	System prompt bypass mumkin

AI Kelajak Yo'nalishi

Xususiyat	Muhimlik	Qiyinlik
Haqiqiy Gemini streaming (ReadableStream)	Kritik	Yuqori
Rate limiting (per user/day)	Kritik	O'rta
Token usage tracking	Kritik	Past
Context window (sliding window)	Yuqori	O'rta
AI memory (pgvector embeddings)	Yuqori	Yuqori
File attachments (PDF, DOCX)	O'rta	O'rta
Voice input (Whisper API)	O'rta	Yuqori
Image analysis (Gemini Vision)	O'rta	O'rta
AI course recommendations	O'rta	Yuqori

6. XAVFSIZLIK AUDITI

Xavfsizlik Bahosi



Kritik Xavfsizlik Muammolari

1. Role Self-Selection — KRITIK

Foydalanuvchi ro'yxatdan o'tishda `role: 'teacher'` yoki `role: 'admin'` yuborishi mumkin. Backend bu qiymatni tekshirmaydi. Bu katta xavfsizlik zaifligi.

Yechim: Public registration'da faqat Student roli. Teacher va admin — faqat admin tomonidan yaratiladi.

2. Email Tekshiruvi Yo'q

Tasdiqlashsiz hisob yaratish mumkin. Spam va abuse xavfi.

Yechim: Supabase Dashboard'dan email verification yoqish (1 daqiqa).

3. Rate Limiting Yo'q (AI Edge Function)

Bir foydalanuvchi cheklovsiz AI so'rov yuborishi mumkin. Butun Gemini API kvotasi tugaydi.

4. AI Prompt Injection

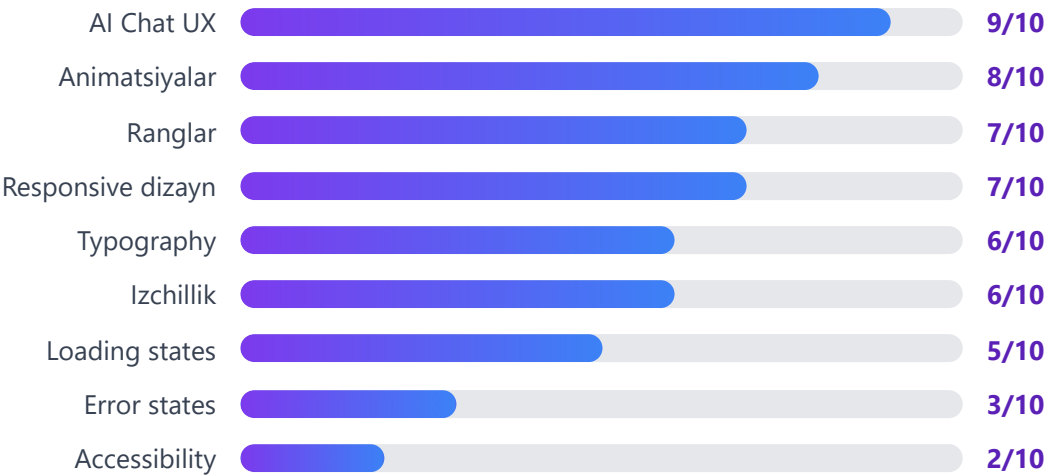
Foydalanuvchi maxsus so'rovlar bilan system prompt'ni bypass qilishi mumkin.

● 5. Faqat Client-Side Fayl Tekshiruvi

Avatar yuklash faqat browser'da tekshiriladi. Server-side validation yo'q.

7. UI/UX AUDITI

Kategoriyalar Bo'yicha Baho



● Accessibility — Kritik Muvaffaqiyatsizlik (2/10)

- Icon-only buttonlarda ARIA label yo'q
- Keyboard navigation testlanmagan
- Screen reader support yo'q
- Color contrast auditlanmagan
- Modal'larda focus trap yo'q
- Skip navigation yo'q

Ko'plab yurisdiksiyalarda ta'lim platformalari uchun accessibility qonuniy talab!

● Brand Izchilsizligi

Platforma ikki nom bilan yuritilmoqda:

- Sidebar/Navbar: **"YordamchiAI"**
- AI Chat: **"Asomiddin AI"**

Bu foydalanuvchida chalkashlik yaratadi. Kelajak White Label uchun brand konfiguratsiyadan kelishi kerak.

8. KELAJAK ARXITEKTURASI

Tavsiya Etiladigan Papka Tuzilmasi

```
src/
├─ app/           ← Router, Providers
├─ config/        ← brand.ts, env.ts, routes.ts, feature-flags.ts
├─ features/      ← ASOSIY O'ZGARISH: Feature modules
│   ├─ auth/
│   ├─ identity/
│   ├─ ai-assistant/
│   │   ├─ components/ ← ChatInput, ChatMessages, MessageBubble...
│   │   ├─ context/    ← ChatContext
│   │   ├─ hooks/      ← useChat, useStreaming
│   │   ├─ pages/      ← thin shell (<100 qator)
│   │   └─ services/   ← ai-chat.service
│   ├─ lessons/
│   ├─ attendance/
│   ├─ tests/
│   ├─ achievements/
│   ├─ homework/      ← Kelajak
│   ├─ notifications/ ← Kelajak
│   └─ payments/      ← Kelajak
├─ shared/         ← Cross-feature reusable
│   ├─ components/ui/ ← Button, Input, Modal, Toast, DataTable...
│   ├─ hooks/        ← useDebounce, useLocalStorage...
│   └─ utils/        ← date.ts, string.ts, format.ts
├─ lib/            ← supabase.ts, utils.ts
└─ styles/         ← globals.css, design-tokens.css
```

Brand Konfiguratsiyasi (White Label)

```
// src/config/brand.ts
export const BRAND = {
  platform: {
    name: process.env.VITE_PLATFORM_NAME ?? 'YordamchiAI',
    tagline: process.env.VITE_PLATFORM_TAGLINE ?? 'AI-powered Learning',
    logo: process.env.VITE_PLATFORM_LOGO ?? '/logo.svg',
  },
  ai: {
```

```
name:      process.env.VITE_AI_NAME    ?? 'AI Yordamchi',
tagline:    process.env.VITE_AI_TAGLINE ?? 'Powered by Gemini 2.5 Flash',
avatarPath: process.env.VITE_AI_AVATAR ?? '/ai-avatar.jpg',
model:      process.env.VITE_AI_MODEL  ?? 'gemini-2.5-flash',
maxTokens:  parseInt(process.env.VITE_AI_MAX_TOKENS ?? '1500'),
},
colors: {
  primary:    process.env.VITE_COLOR_PRIMARY    ?? '#7c3aed',
  secondary:  process.env.VITE_COLOR_SECONDARY  ?? '#4f46e5',
},
} as const
```

Bu tuzilma har bir mijoz uchun alohida `.env` fayli orqali to'liq White Label qilish imkonini beradi.

State Management

Bosqich	Texnologiya	Sabab
Hozir (1-bosqich)	React Context	Maqbul, o'zgarish kerak emas
3 oy (2-bosqich)	+ Zustand	Chat state + identity state uchun
6 oy (3-bosqich)	+ TanStack Query	Server state caching

9. RIVOJLANISH ROADMAP

● BOSQICH 1 — Barqarorlashtirish (1–3 hafta)

Maqsad: Mavjud xususiyatlarni production-ready qilish. Yangi xususiyatlar emas.

Vazifa	Muhimlik	Qiyinlik
Role self-selection'ni olib tashlash	● Kritik	Kichik
avatars storage bucket yaratish	● Kritik	Juda kichik
ErrorBoundary barcha layoutlarga	● Kritik	Kichik
ProfileContext → barcha sahifalar	● Kritik	O'rta
Email verification yoqish	● Kritik	Juda kichik
Kod splitting (React.lazy)	● Kritik	O'rta
AI Rate limiting	● Kritik	O'rta
profiles.updated_at migration	● Kritik	Juda kichik
brand.ts config yaratish	● Yuqori	O'rta
Toast notification system	● Yuqori	O'rta

● BOSQICH 2 — Asosiy Xususiyatlar (4–8 hafta)

Maqsad: Mavjud modullarni yakunlash. Haqiqiy streaming. Homework.

Vazifa	Hafta
--------	-------

Haqiqiy Gemini streaming (ReadableStream)	4
Token usage tracking	4
AIAssistantPage → feature module	4–5
Homework moduli (DB + UI)	5–6
Notifications tizimi	5–6
Certificate DB jadvali + verification	6
Leaderboard sahifasi	6
Student Dashboard real data	5
Playwright test suite	7–8
Accessibility birinchi pass	8

BOSQICH 3 — Skalabillik (9–14 hafta)

Maqsad: Feature module arxitekturasini. Zustand. Analytics.

- Feature-based papka tuzilmasiga migratsiya
- Zustand (chat + identity state)
- Analytics dashboard (Admin)
- AI conversation memory (pgvector)
- Fayl attachments AI chat'da
- Design token system (CSS custom properties)
- DataTable shared component

BOSQICH 4 — Monetizatsiya (15–20 hafta)

Maqsad: To'lov infratuzilmasi. Marketplace. Subscriptions.

- Payments (Stripe/Payme/Click integratsiya)
- Subscription planlar
- Course marketplace
- Teacher revenue sharing

- White label configuration UI

● BOSQICH 5 — Mobile va API (6–12 oy)

Maqsad: React Native app. Public API. Enterprise integrations.

- React Native app (Expo)
- Public REST API
- AI Voice input (Whisper)
- AI Image analysis (Gemini Vision)
- Multi-tenant support
- SSO integratsiya

10. TOP 20 ENG MUHIM YAXSHILANISHLAR

1

Role self-selection'ni olib tashlash

● Kritik xavfsizlik zaifligi — foydalanuvchi o'zi admin bo'la oladi

2

avatars Storage bucket yaratish

● Hozir avatar upload 404 qaytaradi — to'liq ishlamaydi

3

ErrorBoundary barcha layoutlarga

● Bitta JavaScript xatosi butun ilovani to'xtatadi

4

ProfileContext barcha sahifalarda

● Qurilgan lekin hech qayerda ishlatilmaydi — isrof

5

Email verification yoqish

● Tasdiqlashsiz hisob — spam va abuse xavfi

6

Kod splitting (React.lazy)

● 1.37MB bitta bundle — hamma narsani yuklab oladi

7

AI Edge Function rate limiting

● Bir user butun Gemini kvotasini tugata oladi

8

Token usage tracking

● AI xarajatini nazorat qilib bo'lmaydi — moliyaviy xavf

9

/reset-password route yaratish

● Parol tiklash oqimi to'liq emas

10

UserAvatar imgOk re-render bug

● Infinite flicker loop xavfi — avatar yuklanmasa

11

Haqiqiy Gemini streaming

● Hozir setTimeout simulyatsiya — texnik qarz

12

AIAssistantPage feature module'ga bo'lish

● 700+ qatorlik fayl — SRP buzilishi

13**Toast notification system**

● Global feedback yo'q — UX muammosi

14**Soft delete barcha jadvallarda**

● Ma'lumot yo'qolish xavfi

15**profiles.updated_at column**

● Profil o'zgarish tarixi yo'q

16**Homework moduli**

● Ta'lim platformasi uchun zarur — DB ham yo'q

17**Certificate DB jadvali**

● Hozir faqat client-side PDF — tekshirib bo'lmaydi

18**Notifications tizimi**

● Asosiy kommunikatsiya kanali yo'q

19**brand.ts config White Label uchun**

● Har nom hardcoded — White Label imkonsiz

20**Accessibility birinchi pass**

● ARIA labels, keyboard nav — qonuniy talab

11. YAKUNIY BAHO

6/10 Arxitektura	7/10 Database	7/10 UI Sifati
5/10 UX Sifati	4/10 Performance	5/10 Skalabillik
5/10 Maintainability	6/10 Xavfsizlik	7/10 AI Tayyorligi

5.3 / 10

UMUMIY BAHO

Platform MVP bosqichida. Production-ready emas, lekin kuchli poydevorga ega.
Yuqoridagi 1–10 raqamli kritik muammolarni hal qilish — production'ga tayyorlikning 80%'ini beradi.

